



[REDACTED]
DEPARTMENT OF THE NAVY
DIRECTOR, STRATEGIC SYSTEMS PROGRAMS
1210TH STREET SE SUITE 3600
WASHINGTON NAVY YARD DC 20374-5127

SPJA23-51
Code: SPN602

JUSTIFICATION AND APPROVAL (J&A)
FOR USE OF OTHER THAN FULL AND OPEN COMPETITION

1. Contracting Activity. Department of the Navy's Strategic Systems Programs Office (SSP) located at the Washington Navy Yard, 1250 10th Street SE, Washington D.C. 20374.

2. Description of the Action Being Approved. This Justification and Approval (J&A) is for award of a contract on a sole source basis to Charles Stark Draper Laboratory (Draper) for engineering, research and development (R&D), test and analysis, and integration support capabilities in the areas of navigation technology research, inertial measurement, flight test & simulation breakdown and application, and advanced and emerging technology analysis. These efforts are in support of the Navy Conventional Prompt Strike (CPS) hypersonic weapon guidance system for Fiscal Years (FY) [REDACTED].

3. Description of Supplies/Services. Draper will be required to develop specialized engineering efforts and conduct analysis for an advanced Inertial Measurement Unit (IMU) (also referred to as "Hyperflight") flight-testing support, Global Positioning System (GPS) applications, and Electro-Optical (EO) Alternative Navigation (also referred to as "Hypersight") guidance and reentry subsystems. The anticipated contract scope includes:

- (1) Providing advanced engineering development in support of navigation software support and integration, alternative guidance and navigation research, and advanced capability research. These efforts will be applied to maturing navigation aids, sensors, and inertial measurement capabilities for CPS.
- (2) Supporting sequential, progressive testing in support of Multi-service Advanced Capability Hypersonic Test Bed (MACH-TB) and Joint Flight Campaign (JFC) test events as part of continued development and improvement of the CPS Weapon System (WS) capabilities.
- (3) Conducting R&D to create advanced navigation technologies that do not rely on GPS in a contested environment including, but not limited to, EO Alternative Navigation.
- (4) Designing, developing, delivering, and supporting the integration of non-GPS dependent navigation solutions, including EO Alternative Navigation-based navigation and advanced IMU capabilities that optimize the systems for size, weight, and power while maintaining CPS accuracy requirements. Analyzing, designing, delivering and testing iterations of alternate navigation aids and sensors to facilitate non-GPS dependent navigation accuracy.



- (5) Providing testing, analysis, and integration support for common Software In The Loop (SIL) and Hardware In the Loop (HIL). Conduct post-flight analysis of navigation performance in support of MACH-TB and JFC test events.
- (6) Performing technical trade studies to assess the required capabilities for fusing multiple navigation signal inputs that improve system performance and probability of arrival at designated targets.

The anticipated contract will have a period of performance from [REDACTED] and continue through [REDACTED]. The estimated dollar value of this contract is [REDACTED] and includes the following funding amounts:

[REDACTED]

Note: The contract will include contract text permitting an optional increase of not to exceed 30 percent of the total value associated with each LOE contract line item to accommodate emergent in-scope requirements. This increase is not accounted for in the above table.

4. Statutory Authority Permitting Other Than Full and Open Competition. 10 U.S.C. 3204(a)(1), only one responsible source and no other supplies or services will satisfy agency requirements.

5. Rationale Justifying use of Cited Authority. Pursuant to 10 U.S.C. 3204(a)(1), as implemented by FAR 6.302-1(a)(2)(ii), the supplies and services needed by the agency for this follow-on contract are available only from Draper, and no other type of supplies or services will satisfy the needs of the agency. Draper is the only responsible source available to provide the highly specialized R&D and advanced engineering design of the EO Alternative Navigation and IMU capabilities to support hypersonic weapon guidance requirements. Draper is the historical design agent and Subject Matter Expert (SME) for numerous navigation capabilities across several guidance systems for different Navy missiles and is the only contractor to have full access to the design, development, and production of all subcomponents through the various generations of fielded missile systems. The diverse level of experience allows Draper to leverage its knowledge of SSP R&D initiatives, as well as the navigation requirements and capabilities of multiple, distinct weapon systems, all to the benefit of the CPS Program Office. Draper is the lead source of the CPS Inertial Navigation and Guidance Subsystems and the EO Alternative Navigation technology is currently being designed and developed by Draper under the existing FY22 OMNIBUS contract (SSP Contract No.: N00030-22-C-6002).

Additionally, Draper is the only source capable of providing Guidance, Navigation, and Control (GNC) designs for the CPS Hypersonic WS, GNC software source code, and performing detailed analysis of all factory and fleet performance data. Draper also possesses the engineering design expertise to manage the development for the entirety of EO Alternative Navigation and IMU.

Without access to Draper's current facilities and GNC expertise, assigning the effort to another contractor other than Draper would seriously impact the CPS development effort since a non-GPS dependent navigation capability is a critical objective of CPS.

Awarding to any source other than Draper would result in unacceptable delays in fulfilling the agency's requirements, specifically, [REDACTED] in fielding the CPS weapon system. Ultimately, this would result in a failure to meet the operational need for a hypersonic capability as outlined in the 2018 National Defense Strategy. This critical weapon system is needed to counter peer and near-peer adversaries' development of a suite of fires capabilities, supported by an integrated mix of sensors, surface-to-air, and surface-to-surface shooters, threatening the U.S. joint force domination.

The estimated timeline for the [REDACTED] is based on multiple factors. Draper's facility investments in the GPS, system test areas, and program involvement are directly applicable to engineering efforts that are unavailable from any other organization. An alternate source would have to duplicate the capabilities of the existing Draper facility, establish the infrastructure, and install equipment required to meet SSP engineering, quality, and reliability requirements. The estimated [REDACTED] also considers additional time required to hire and train the contractor workforce to duplicate Draper's capabilities and the associated learning curve to meet SSP engineering, quality, and reliability requirements.

Draper has continuously executed quality control, engineering, and reliability disciplines required by CPS. Draper is a non-profit laboratory, with access to proprietary information from industry partners, allowing it to tailor a design solution, which optimizes performance for CPS. Draper is also the only qualified contractor with the ability to execute the Government's requirements to support the continued development of navigation solutions for CPS. Draper's experience with alternative navigation solutions will enable CPS to continue developing alternative GNC solutions without delaying the program. Utilizing an alternate source would introduce unknown design risks and unproven capability that would cause major impacts to the overall CPS program schedule.

6. Description of Efforts Made to Solicit Offers from as Many Respondents as Practicable.

- a. A Sources Sought Notice (SSN) was posted to the System for Award Management (SAM) website on 27 July 2023 for 20 days, and this posting received six responses. Of these responses, only one respondent, Draper, clearly indicated the ability to fulfill the Government's needs in its entirety. The Government technical SME for this procurement assessed that no respondents beyond Draper could assume the support requirements without introducing unacceptable delays and schedule risks to the program. Other than Draper, the following sources expressed interest in this requirement: [REDACTED]
[REDACTED]
[REDACTED] While [REDACTED] and [REDACTED] expressed interest, they did not submit a formal capability statement in response to the SSN, and notwithstanding that affirmative expressions of interest were received from [REDACTED], it was determined that none of these sources possess the capability to execute the CPS program requirements without unacceptable delays.

- b. In accordance with FAR 5.202, the Government issued a pre-solicitation notice on November 30, 2023, with a response deadline of December 14, 2023. No responses were received.
- c. Other market research efforts have been ongoing to explore potential solutions to the Government's requirement for specialized engineering for advanced research and development support, simulation, testing, and analysis in the field of GNC and GNC solutions that do not rely on GPS. These efforts include contacting SMEs in the Government National Team to help assess industry capabilities regarding the Government's requirement. SSP also reviewed the results of recent market research of similar procurements and further examined systems such as SAM, the Federal Procurement Data System, and the Small Business Administration's Dynamic Small Business Search to seek entities that had previously fulfilled related procurement requirements. At this time, the market research confirms that it is not practical, for the reasons discussed in paragraph 5 above, for any company other than Draper to provide the required services and supplies given the complexity of the efforts being procured in the highly specialized and unique CPS Hypersonic WS without introducing schedule risks to the program. While no other sources have been identified to provide prime contractor support, Draper will be required to promote competition where opportunities exist in selecting subcontractors and suppliers.

7. Determination of Fair and Reasonable Cost. The Contracting Officer has determined the anticipated cost to the Government of the supplies and services covered by this J&A will be fair and reasonable.

8. Actions to Remove Barriers to Future Competition. For the reasons set forth in paragraph 5 above, SSP has no plans to compete a future contract for the types of supplies and services covered by this J&A. If another potential source emerges through the continued market research processes, SSP will assess whether competition for future requirements is feasible, while analyzing any potential impact to cost, schedule, and program risk. Additionally, the prime contractor for this effort will utilize competition amongst suppliers and subcontracting efforts where feasible. It is noted that the current design, development, and delivery of this requirement does not exclude any company from participation in downstream subcontracting efforts. The Government will take delivery of a Technical Data Package in [REDACTED] that will promote competition in the future.

As this is a follow-on for supplies and services previously awarded on a non-competitive basis, in accordance with DFARS PGI 206.304(a)(S-70)(i), the prior J&A was provided to the approval authority for review. Approval of this J&A signifies the approval authority has made the determination required by DFARS PGI 206.304(a)(S-70)(ii).

CERTIFICATIONS AND APPROVAL

TECHNICAL/REQUIREMENTS CERTIFICATION

I certify that the facts and representations under my cognizance which are included in this Justification and its supporting acquisition planning documents, except as noted herein are complete and accurate to the best of my knowledge and belief.

TECHNICAL COGNIZANCE:

[REDACTED]

REQUIREMENTS COGNIZANCE:

[REDACTED]

[REDACTED]

CHIEF OF CONTRACTING OFFICE CERTIFICATION:

I certify that this Justification is accurate and complete to the best of my knowledge and belief.

[REDACTED]

SENIOR PROCUREMENT EXECUTIVE APPROVAL

Upon the basis of the above Justification, I hereby approve, as Senior Procurement Executive of the Navy, the solicitation of the proposed procurements described herein using other than full and open competition, pursuant to the authority of 10 U.S.C. 3204(a)(1), Only one responsible source and no other supplies or services will satisfy agency requirements.

APPROVED:

